

Correlation and Regression

Correlation

Correlation refers to the relationship between two variables. It tells us how one variable changes when the other variable changes.

For example: If we study the relationship between hours studied and marks scored, correlation helps us understand whether students who study more generally score more or less.

- If both variables increase or decrease together, the correlation is positive. Example: As hours of study increases, marks also increase.

- If one variable increases while the other decreases, the correlation is negative. Example: As the number of days absent increases, marks decrease.

Coefficient of Correlation or Pearson's Correlation Coefficient

The coefficient of correlation is a numerical measure of the strength and direction of the relationship between two variables.

- It is denoted by $r(X, Y)$.

$r(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$, where σ_X is the standard deviation of X and σ_Y is the standard deviation of Y .

$\text{Cov}(X, Y) = \mu_{11} = E[\{X - E(X)\}\{Y - E(Y)\}] = E(XY) - E(X)E(Y)$

$r(X, Y)$ varies between -1 and 1.

Two independent variables are uncorrelated, i.e., if X and Y are independent variables, then $r(X, Y) = 0$.

Examples

1. Calculate the correlation coefficient (r) between X and Y .

X	4	6	8	10	12	14	16
Y	10	12	15	18	20	24	28

2. Find the value of r for the following data:

X	3	5	7	9	11	13	15	17
Y	20	18	16	15	14	12	10	8

3. Compute the coefficient of correlation (r):

X	2	4	5	6	8	9	11
Y	7	9	10	13	15	18	20

4. Calculate the correlation coefficient (r) from the following data:

X	10	12	14	16	18	22	25	28
Y	30	28	26	25	24	22	21	20

5. Find r for the following paired data:

X	5	7	9	10	11	14	15
Y	50	55	60	63	65	70	75